

Mini Project: Supermarket Sales Analysis

Project Overview

This project performs **data analysis on supermarket sales data** to uncover patterns in customer behavior, product performance, and sales trends. The analysis includes data preprocessing, exploratory data analysis (EDA), visualization, and insight generation using Python.

Objectives

- Load and understand the dataset
- Clean data
- Perform Exploratory Data Analysis (EDA)
- Visualize trends and patterns
- Derive meaningful business insights

Summary & Key Insights

- **Food and Beverages** generate the highest overall sales among all product lines.
- **Member customers** spend more on average than normal customers.
- **E-wallet** is the most frequently used payment method.
- Sales are fairly consistent across dates with slight weekend peaks.
- Female customers show slightly **higher median spending** than male customers.
- Strong correlation exists between **Unit price, Quantity, and Total sales**.
- Cash payments are less popular compared to digital methods.
- Certain product lines show higher variability in spending behavior.

Conclusion

This supermarket sales analysis highlights key factors influencing revenue such as product lines, customer types, and payment preferences. The insights can help supermarket managers:

- Optimize inventory for high-performing product lines
- Encourage membership programs
- Promote digital payment methods